THE MAGNETIC PROPERTIES OF THE THOLEIITE DYKES OF NORTH ENGLAND

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(Communicated by S. Chapman, F.R.S.)

(Received 1948 June 30)

Summary

The magnetic profiles taken over various members of the tholeiite dykes of the North of England show that they possess abnormal magnetic polarization. Direct measurement of the residual magnetism gives a series of magnetic vectors which could arise from a magnetizing field in opposition to the present field. Minor deviations in direction can be explained by flow of the molten material below the Curie point of the magnetic content.

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1. *Introduction.*—During the summer of 1946 it was found that the magnetic anomaly in the Earth's vertical field in the vicinity of the Armathwaite dyke in Cumberland differed appreciably from that to be expected on the basis of induced magnetism. The initial examination covered a length of about three miles of the intrusion, and at all points the abnormal conditions were obtained. A later examination demonstrated that the peculiarities persisted for at least an additional 13 miles, the observations, together with scattered outcrops, leading to a detailed and accurate map of the igneous body which is obscured by a layer of glacial drift over the surface. As the Armathwaite dyke, and its easterly extension in the Cleveland dyke, is one of a suit of dykes, all of similar age, the investigations were extended to cover the other main members, four in all, at a small number of widely separated points. The field survey, and the results of laboratory investigations arising from it, form the subject of this paper.

2. *The dyke system.*—The positions of the five dykes are shown in Fig. 1. The parts examined lie within a roughly triangular area with its apex near the Firth of Clyde and its 50-mile base on the East Coast of England about 140 miles distant. Locally the names of the dykes vary, as for example the Armathwaite dyke which forms a short section of the most southerly member. Here they will be called, from north to south, the Acklington, the Morpeth, the Barrmill, the Eskdalemuir and the Cleveland dykes respectively. As will be seen, the dykes all show similar strikes ranging from north–west, to a few degrees north of west. All the members, which are of andesite, are of Tertiary age, their injection occurring some 30 million years ago, and their petrological characteristics are very similar. Although there are considerable breaks between certain sections, on geological grounds there can be little doubt about their continuity and correct correlation (1). Very few exposures of the Morpeth dyke exist, but as this one is the only olivine-bearing member of the series it is readily identified. Although there are local changes the dykes are nearly vertical for most of their lengths and their widths vary from 30 to 80 ft. One point of major importance

in the present discussion is that, since their formation, there have been no significant earth movements, so that their present position and attitude has been maintained throughout their life.

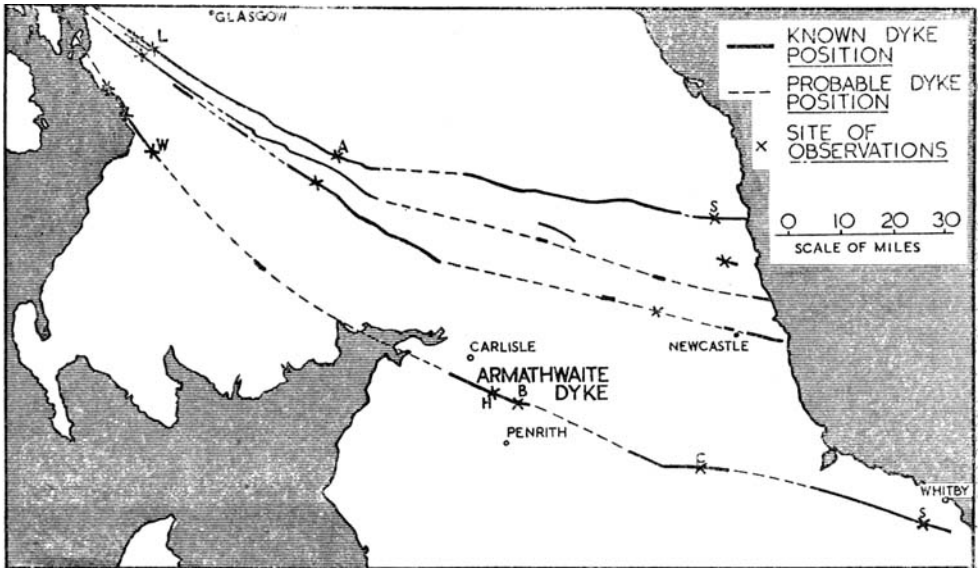


FIG. 1.—*Map of tholeiite dykes showing location of measurements.*

L: *Lugton*	H: *Hazel Cottage*
A: *Abington*	B: *Blunderfield*
S: *Swarland*	C: *Cockfield*
W: *Wallace's Seat*	S: *Sleight's Moor*

3. *General procedure.*—As is well known, the interpretation of magnetic anomalies is not unique, and so the data supplied by the measurement of the magnetic field near a body is insufficient to yield in an unambiguous manner its magnetic state. Accordingly, where conditions permitted, in addition to the field observations, a set of oriented rock specimens was collected from the dyke. They were taken along a line perpendicular to the dyke and at known distances from one of its contacts with the country rock. In the laboratory, measurements were made of the magnetic susceptibility and of the magnitude and direction of any permanent residual intensity of magnetization possessed by the specimen. During the observations the magnetic fields to which the specimen was exposed differed very little from the Earth's field and so the properties were determined under conditions corresponding to those in its natural state. The instrument used for the laboratory determinations allowed the susceptibility to be measured to the nearest 10×10^{-6} c.g.s. units and the residual polarizations to a similar accuracy. The permanent polarization may have suffered, due to the mechanical shock of removing the specimen and also the subsequent cutting to size by a diamond saw, but direct observation on the latter influence demonstrated a change of a few per cent only.

From the measured properties, the resultant intensity of magnetization across the dyke could be estimated. For this purpose the vertical field was taken as 0·45 gauss and the horizontal component 0·17 gauss, making an angle of 348° (12° W.) with the geographical meridian. The magnetic anomaly was then computed and compared with the field observations, the comparison showing whether

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the samples of rock could be taken as magnetically representative of the dyke material.

4. *Position of the field observations.*—The greatest number of measurements were made over the 16-mile section of the Cleveland dyke near Armathwaite, which was examined by observations along 52 traverses across the feature. By combining these results with the relatively small number of sporadic outcrops, the course of the dyke is known in considerable detail. It has not proved to be a continuous body as suggested by the continuous line in Fig. 1, but is revealed as a number of short sections varying in length from 3000 to 10,000 ft. The sections are usually offset slightly relative to each other but they maintain the same general strike direction. In many cases the sections overlap, while in two places the main dyke divides into a complex of subparallel dykes.

In the regional survey, efforts were made to select sites which satisfied certain conditions. It was necessary for the dyke to be well exposed, so that a complete set of specimens was available, and that the geometry of the dyke and thickness of cover should be known. Further, to avoid complications it was desirable that the surrounding rocks should be non-magnetic. Fifteen locations were chosen, by reference to the appropriate Geological Memoirs but, as the mapping of the area was done at the end of the last century, the actual conditions were not always satisfactory. As a result, only seven complete sets of specimens were obtained, three from the Acklington dyke at Lugton, Abington and Swarland, and four from the Cleveland dyke at Wallace's Seat, Hazel Cottage, Blunderfield quarry and Cockfield. At other places it was only possible to obtain the field curve and an isolated specimen from a small exposure. It may be noted here that the magnetic anomalies at Beith and Morpeth on the Morpeth dyke and at Abington on the Eskdalemuir dyke were found some distance (up to 350 ft.) from the position marked on the geological map. In all, then, in addition to the seven locations from which specimens were obtained, 59 field curves were also available for analysis.

5. *The field results.*—For a vertical dyke running east–west in England, if the intensity of magnetization is due to induction in the Earth's field, south polarity would be found on the top and on the southern contact and north polarity on the northern contact. Since a buried south pole will increase the downward vertical component, the resulting anomaly in this component will show an increase over the southern wall of the dyke and a smaller decrease near the northern contact. As the strike of the dyke changes to north–south, the transverse intensity of magnetization is reduced and finally a symmetrical positive anomaly is left, due to the vertical magnetization only. It can be shown that in any uniformly magnetized dyke only one maximum and one minimum in the vertical anomaly can occur.

Of all the field curves measured, only six showed curves compatible with the effects of induction. One of these occurred at Sleight's Moor at the eastern end of the Cleveland dyke and the other five on consecutive traverses in the Armathwaite area of the detailed survey on the Cleveland dyke, and these were all related to the same unbroken length of the dyke. Beyond the gaps at the ends of this length the curves correspond to all the other cases in which the anomalies were dominantly negative, but of varied character. A selection of these is shown by the continuous curves in Figs. 2 and 3 in which all, with the exception of the normal shape obtained at Sleight's Moor, refer to areas in which a set of specimens

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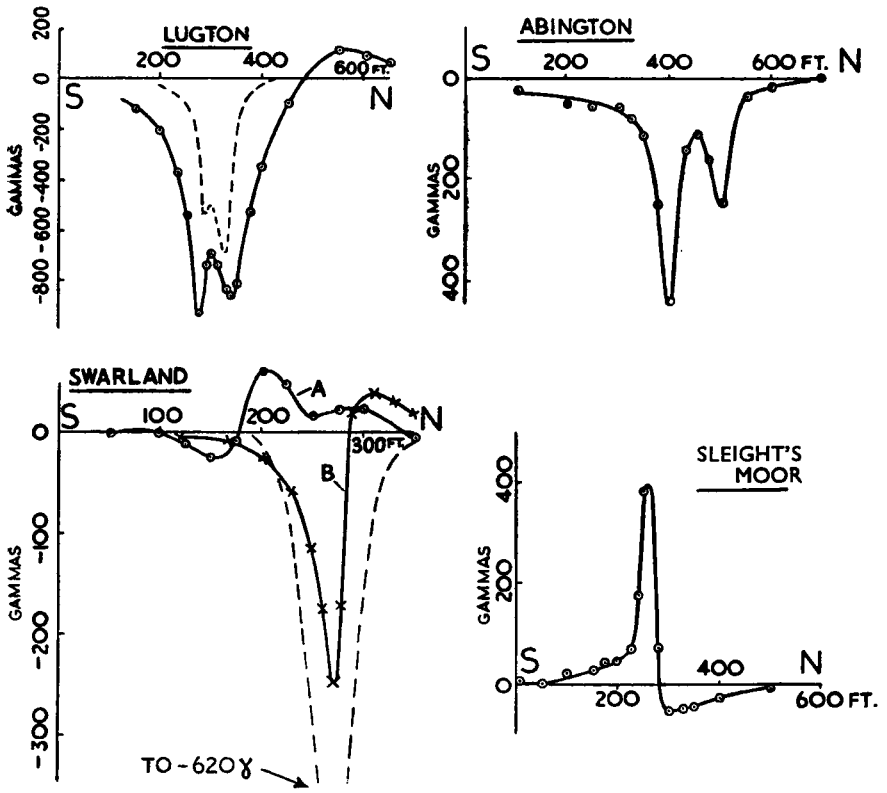


FIG. 2.—*Measured and calculated magnetic anomalies over the Acklington an Cleveland dykes.*

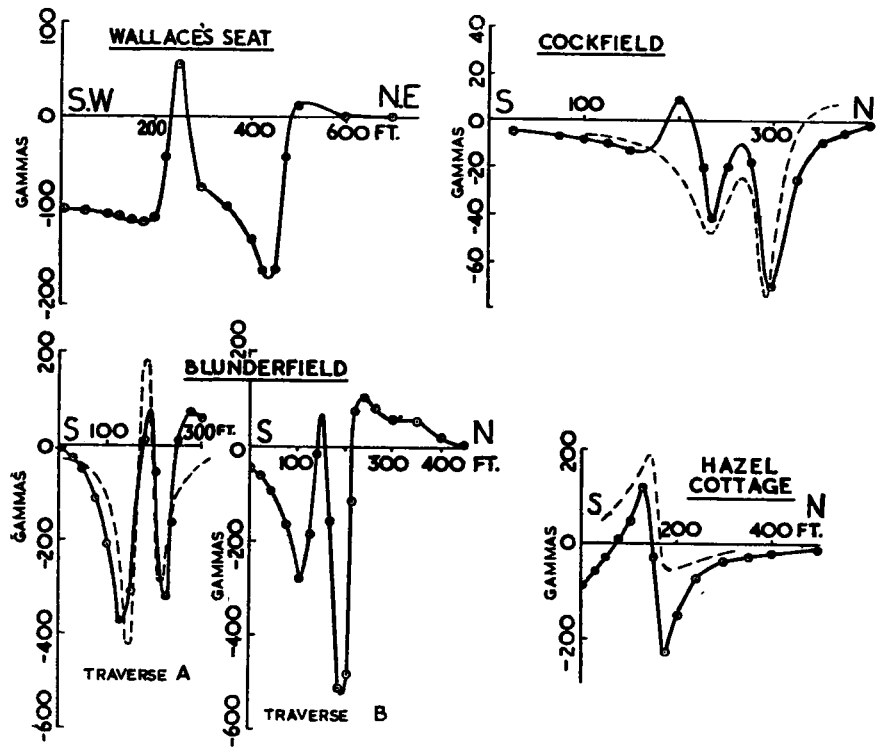


FIG. 3.—*Measured and calculated anomalies over the Cleveland dyke.*

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was collected. All the traverses were approximately at right angles to the length of the dyke and the graphs give the field variation from the south end of the line towards the north.

Certain of these suggest that the known dyke is not the only body producing the anomaly, as at the Abington exposure, at Wallace's Seat and at Hazel Cottage. In the first of these, the smaller sharp minimum about 100 ft. to the north of the main anomaly could not arise from the narrow 30-ft. wide dyke at a depth of 10 ft. from the surface. At Wallace's Seat the curve is unusual, but it can be tentatively related to the known geology. The geological map (Ayrshire 6″ Sheet XXXIII, NE.) shows a sharp deviation of the dyke joining two sections of the same strike direction. Each anomaly lies on the line of one section, suggesting a break with overlap between the two sections rather than a continuous body. The failure of the curve to return to its base level can be attributed to nearby monchiquite plugs intruded into the Coal Measures here, and which are exposed nearby on the banks of the River Ayr. In the detailed survey at Armathwaite, similar breaks were observed, often accompanied by a change in the shape of the magnetic profile. At Hazel Cottage, the continuous decrease in field to the south must arise from another body unless considerable change in dyke width and other properties occurs at depth.

The two curves from Swarland were over lines 130 ft. apart and demonstrate a considerable change in the magnetic properties along the dyke. Similar results were obtained by Granville Poole, Whetton and Taylor (**2**) in their survey of this area. It is to be noted that the dyke, as revealed by their work, swings to the north, and its position differs appreciably from that inferred from surface evidence only.

The Abington and Swarland exposures (in the latter case curve B is considered, as it was nearer the quarry from which the specimens were taken) show a simple minimum. On the other hand, the Lugton exposure exhibits a double minimum. The central maximum is better developed in the Cockfield exposure and at Blunderfield quarry a central positive anomaly is flanked by two minima. At Hazel Cottage a positive anomaly to the south is followed by a negative one, but the ratio of the two is incompatible with induced magnetism only. Such curves show that the intensity of magnetization differs appreciably from that of the simple case, that in the majority of cases it is mainly vertical with the positive polarity upwards and, since more than one maximum and one minimum occur, that it is not uniform. The rocks accordingly must possess residual magnetism of considerably greater magnitude than, and roughly opposed to, the induced polarization.

6. *The laboratory examination of specimens.*—The results of the laboratory measurements are shown in Figs. 4, 5 and 6. The variations of the susceptibility k, and of the intensity of the total residual magnetization R, with position across the dyke are shown in Figs. 4 and 5, the former giving the results for the Acklington dyke and the latter for the Cleveland dyke. In Fig. 6 the direction of the permanent magnetic vectors are plotted without regard to magnitude; the upper of each pair of diagrams shows the projection of the dip on to the vertical plane containing the traverse and the lower is a plan giving the angle between the direction of the traverse and the azimuth of the vector. In all these diagrams the distance is from the southern edge of the dyke.

It will be seen that the susceptibility and the magnitude of the residual magnetism vary in a rather erratic manner across the structure but, in general, the

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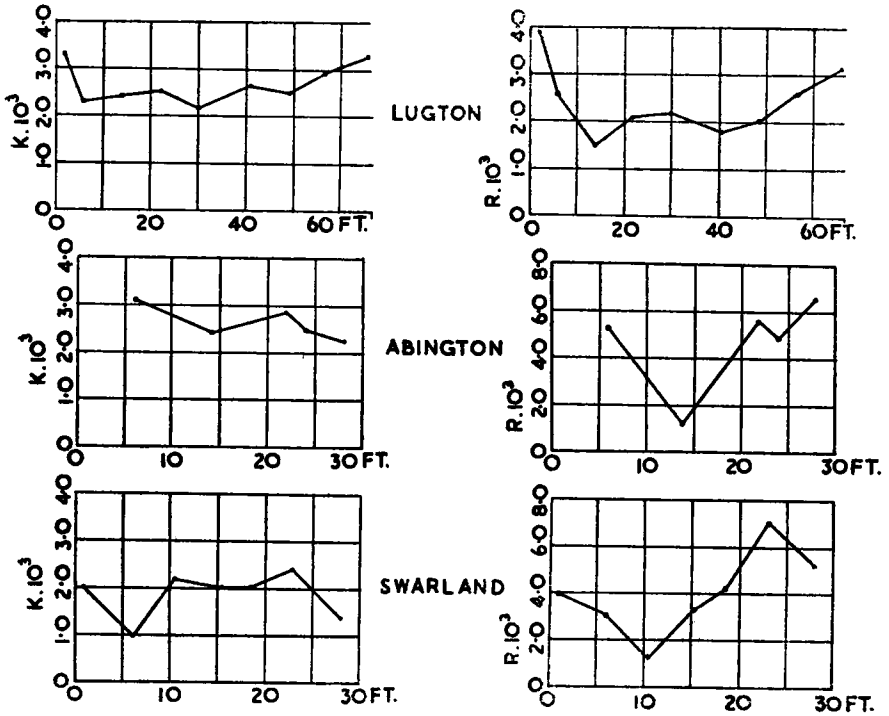


FIG. 4.—*Measured susceptibility and residual intensity of magnetization across the Acklington dyke.*

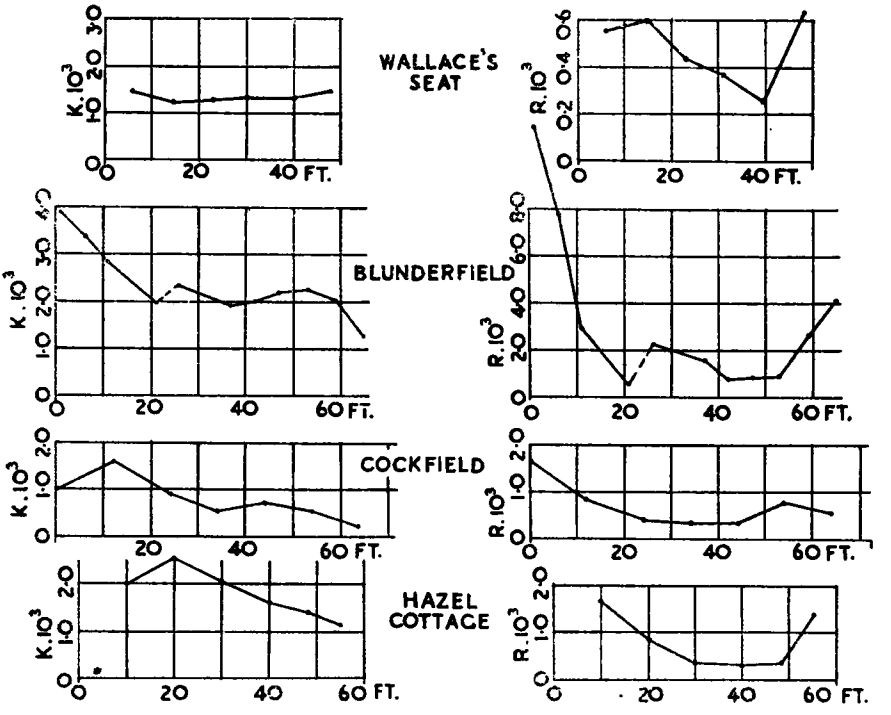


FIG. 5.—*Measured susceptibility and residual intensity of magnetization across the Cleveland dyke.*

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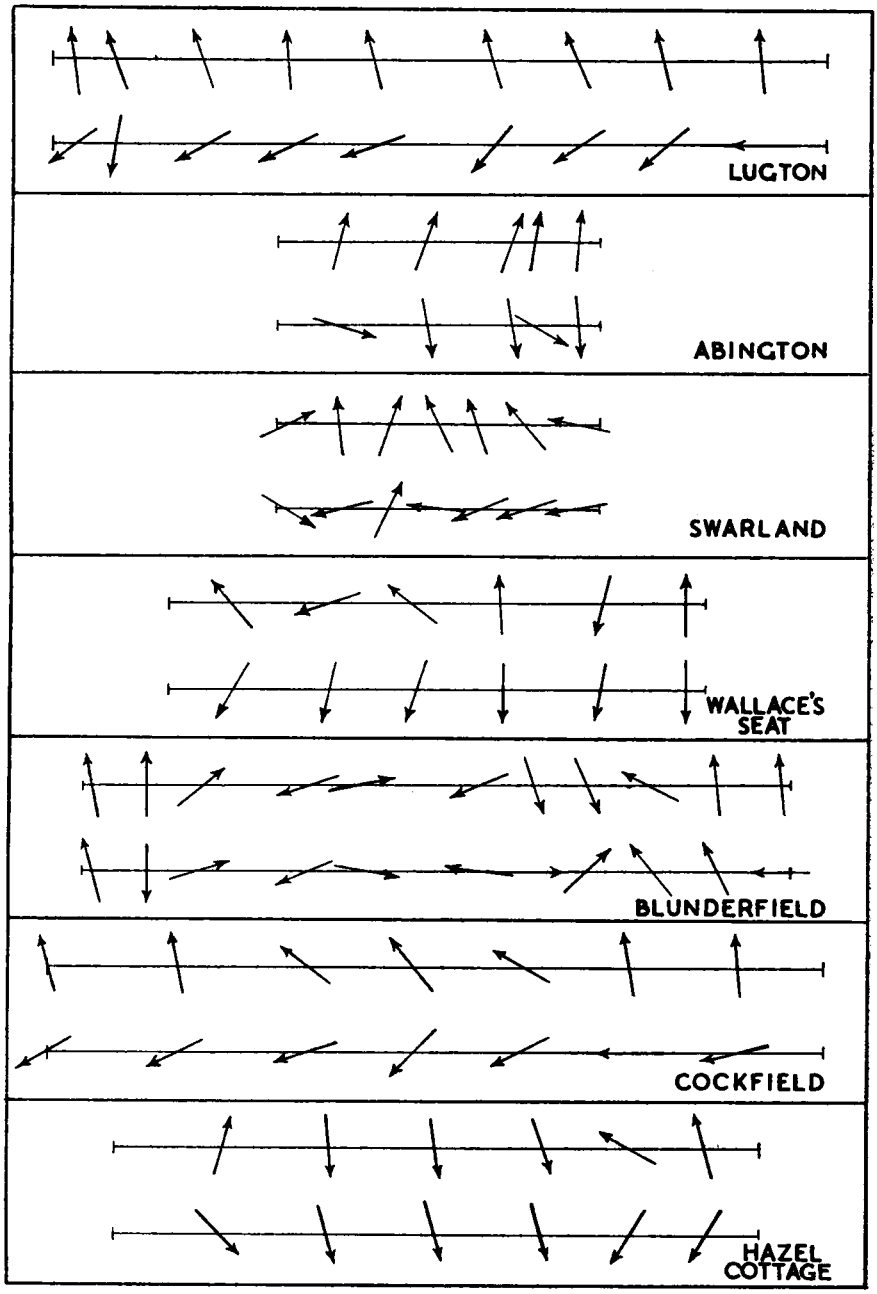


FIG. 6.—*Direction of residual intensity of magnetization.*

residual magnetism is greater on the flanks than in the centre. The variation confirms that the resultant intensity of magnetization is not uniform. The direction of the vectors at Lugton and Abington are extraordinarily uniform, but on the same dyke at the Swarland exposure the directions are not so regular. The Cleveland dyke shows very varied directions at Blunderfield quarry, uniformity of azimuth but not of dip at Wallace's Seat and Hazel Cottage but reasonable agreement between the vectors at the Cockfield exposure. These directions

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are subject to certain errors of observation. In addition to the difficulty of collecting accurately oriented specimens, where the dip is large the small horizontal components cannot be measured with great precision, leading to relatively large errors in the azimuth.

The total intensity of magnetization calculated with these data reveals that, in general, the vertical component is greater than the transverse horizontal component. As a result, and because the transverse component is less effective in modifying the Earth's vertical component, the main features in the field curves can be accounted for by the vertical component of the intensity of magnetization. Thus, at Lugton this component is about 2000×10^{-6} c.g.s. units on the flanks, compared with half this value at the centre. Combined with the shallow depth of the top this results in the two minima. At Cockfield the south and north flanks have values $+200 \times 10^{-6}$ and $+400 \times 10^{-6}$ c.g.s. units respectively while the central 30 ft. have a mean value of -100×10^{-6}, just too small to give a central positive anomaly. At Blunderfield the negative intensity in the centre had a relatively greater value in comparison with the flank values giving the observed central positive anomaly.

The computed curves are shown by broken lines in Figs. 2 and 3. To obtain these, the dyke was divided into a set of narrow vertical strips, in each of which the intensity of magnetization could be regarded as uniform, the vertical and horizontal components of intensity being considered independently. To this extent the calculated values are approximate, but in addition it was assumed that the magnetic properties remained constant along the dyke and also in depth. It is apparent that the rock properties measured in the laboratory are adequate to account for the observations in general shape and order of magnitude. There are certain discrepancies, but these are to be expected. Thus, at Lugton the measured curve falls away more slowly than the calculated curve, a feature which would result if the magnetic properties increased with depth. Such an increase would also improve the agreement in magnitude. Depending on conditions, the traverses were 50 ft. or so away from the exposure over ground covered by drift. Accordingly the precise thickness of cover is unknown and quite small variations in this will influence the magnitude.

At Lugton, Blunderfield quarry and Cockfield the agreement is very satisfactory and the general shape is good at Hazel Cottage. As has been noted, it seems probable that in the last case some other body is also contributing to the field. The comparison at Abington is not so good, the calculated curve (not shown) giving a simple and nearly symmetrical minimum corresponding to the larger anomaly observed, but its magnitude is -1180 gamma. The difference here cannot be fully explained, but the duplication of the anomaly 100 ft. away on a smaller scale suggests a break in the feature with an overlap between sections. Experience at Armathwaite shows that such features are accompanied by changing conditions along the dyke. At Swarland there is direct evidence of such longitudinal variation in magnetic properties. Here a displacement of 120 ft. changes the measured anomaly from negative to positive and so the calculated anomaly of -620 gamma from specimens collected 30 ft. away can be considered reasonable. The calculation has not been made for the data at Wallace's Seat mainly because of the uncertainty in the measured anomaly and the geological complexity there. The curve, however, would be a nearly symmetrical positive anomaly of some 70–80 gamma.

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Apart from this last case, the observed properties account for the peculiarities obtained in the field curves and it may be concluded that the specimens can be regarded as magnetically representative of the dyke material.

7. *Discussion of the results.*—One of the outstanding features of the results is the direction of the permanent magnetic vector. At Lugton, Abington, Swarland and Cockfield they all point upwards, forming a set of roughly parallel vectors, particularly in the first-named exposure. At the other three locations the vectors are more erratic but where the intensity is large, viz. on the flanks, again they point upwards. The direction of the mean vector at each place has been computed, and the results are shown in Fig. 7A. The diagram represents a plan of the surface of a hemisphere, with its pole uppermost, and with circles of equal dip and the azimuths from the north direction marked on it. The direction in any case is given by the line joining the centre of the hemisphere to the points plotted on the surface. It can be concluded that the direction of the residual magnetism is roughly in the opposite direction to the present Earth's field and, taking into account the persistent negative anomaly obtained over the whole of this area, it would appear that these conditions are applicable to the whole.

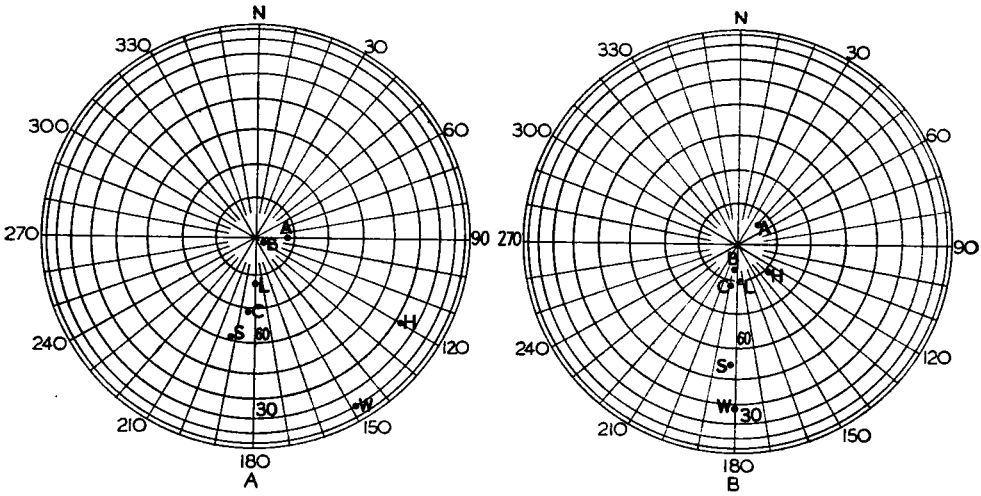


FIG. 7.—*Mean values of the direction of residual magnetism.*

It is usually accepted that the residual magnetism is acquired as the body cools down through the Curie temperature of the magnetic material within the dyke, such magnetism being in the direction of the field acting at that time. Experience in the laboratory, and observations on lava recently extruded at the Earth's surface, confirms this point. Hence, since there has been no significant earth movement in this region during the 30 million years this dyke system has existed, the Earth's field in the area at the time of cooling through the transition temperature would appear to have been approximately in opposition to its present direction. This would account for the conditions observed at Lugton and elsewhere, but the present state of magnetization is more complex than the simple picture suggests and it is necessary to find a mechanism to produce the variations in the direction of the residual intensity.

In general, any theory of magnetization in a uniform field will lead to a set of parallel vectors and hence a rotation of the magnetized material appears a

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necessity. If the dyke material can retain some measure of mobility below the Curie point, then a relatively small movement would be adequate to account for the observed facts. Direct measurement of the transition temperature of specimens from Blunderfield quarry and Lugton gave an average value of 500 deg. C., but on second heating each specimen gave a value 20–30 deg. lower. Chevallier and Pierre (**3**) attribute this change to a partial separation of the magnetite into Fe_2O_3 and FeO, the Curie point for magnetite itself being 580 deg. C. Thus it would appear that on the first cooling *in situ* the magnetic properties would be acquired between 500 deg. C. and 580 deg. C. There is little direct evidence on the mobility of basic rocks at these low temperatures. The Geophysical Laboratory of the Carnegie Institute of Washington has shown that the presence of small amounts of water can reduce the solidifying points of certain minerals, either separately or in pairs, by a very remarkable amount (**4**). Day (**5**) quotes the case in which a basic lava continued to flow under its own weight at 600 deg. C. When the lava solidified a temperature of 1300 deg. C. was required to melt it again, illustrating well the fluxing action of the volatile constituents even at atmospheric pressures. At higher pressures the greater volatile content may well result in mobility down to the temperatures required here.

Flow structures of the type required are well known in igneous rocks (**6**) but are generally more readily appreciated in acid intrusions rather than in basic. A microscopic examination of a number of specimens was made in the hope of detecting evidence of the resulting structure but the results were disappointing. Only in four cases, one each from Swarland, Blunderfield quarry, Cockfield and Hazel Cottage was there any evidence of linear parallelism of the phenocrysts in the sections, and even here it was not strongly developed. Nevertheless, the direction of flow appeared compatible with that necessary to give the magnetic orientations. The lack of support in this direction is not surprising, since flow structure is best appreciated on a large scale rather than by microscopic investigation.

It is impossible to predict the pattern produced by flow in an igneous body, since it will depend on a number of unknown factors, such as the relative viscosity of the parts of the magma involved and the seat of the forces causing the displacement. Quite a simple flow pattern would account for the directions observed at Blunderfield quarry, if it is assumed that specimens 3 and 4 from the south contact are duplicated approximately by specimens 5 and 6. This is quite feasible, as the plan of the quarry face has a Z shape, specimens 1–4 being collected from one face and 5 onwards from the other. In addition, the observed properties of the two pairs are similar. The necessary lines of flow to give the observed orientations are shown in Fig. 8. Naturally, this picture is tentative, and it

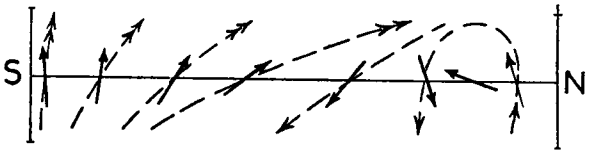


FIG. 8.—*Possible flow to give pattern of vectors at Blunderfield.*

would be necessary to examine the magnetic vectors in the surrounding rocks to confirm it. Such movements would also account for the longitudinal variation as shown by the changing detail of the magnetic profiles. There is some evidence

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to suggest that the Cleveland dyke did not reach the surface when injected. Thus Teall (**7**) and Geikie (**8**) have emphasized the fact that the dykes, west of Dalston Hall, tend to die away without reaching the surface. Accordingly, the inferred flow structure may be due to the fact that erosion has reached the critical level where the top of the dyke is exposed.

If this interpretation is correct then, since the flanks cool most rapidly, the permanent magnetic vectors here should exhibit the least disturbance. Accordingly, there should be better agreement between the mean of the vectors near the flanks. These are plotted in Fig. 7B in which an improved grouping is to be found. The two which appear most exceptional are the results at Swarland (193° azimuth, 61° dip) and Wallace's Seat (180°, 37°), the latter having moved from (120°, 20°) on A. As has been shown, Swarland appears to be a region of great disturbance by flow.

These results agree with the survey carried out by McLintock and Phemister (**9**) over the Lornty dyke in Scotland, a Tertiary dyke of the same group as these investigated here. Of the same approximate age a basalt dyke in Saxony has been shown by Schulze (**10**) to possess abnormal characteristics and Reich (**11**) reports an igneous intrusion in Germany, probably of Tertiary Age, showing anomalous polarization. Similar results have been obtained over the extensive Pilansberg dyke system (**12**) of the Southern Transvaal, an intrusion of Palaeozoic Age and over a diabase dyke in Brazil (**13**). In a private communication Dr Marsden has drawn our attention to a rhyolite intrusion, whose age can be fixed at about 15 million years, which again exhibits adverse magnetization. Thus, such characteristics are fairly common in both surface distribution and in age. There can be no doubt of the changed direction of the magnetic field necessary to produce the observed polarization. Whether such inverted fields were widespread over the Earth's surface in the past, or whether they were an abnormal, but local, state associated with the conditions necessary for the invasion of the crust by molten magmas cannot as yet be decided. It is worth noting that igneous intrusions are known with magnetic anomalies compatible with the present inducing field. Here an investigation of the coercive force may reveal conditions which allow a rapid loss of permanent magnetism.

8. *The magnetic properties of the dyke material.*—The observed values of the susceptibility of the rock material reveal a considerable change both across and along the dykes. This variation would appear to be due, in part, to the changing magnetite content arising from differentiation. No quantitative check has been made on this point but a microscopic examination of thin sections confirms the fact qualitatively. In addition, there would appear to be a change in the properties of the magnetic material itself, since, in all cases, the ratio of permanent intensity to susceptibility is greater on the flanks than at the centre (Fig. 9). The average values are given in Table I. Since the Earth's field is about 0·5 gauss, the ratio of permanent to induced magnetism is equal to twice these values. The residual magnetism is probably lower than the original, since there may have been some demagnetization during the long exposure to the present demagnetizing field, while collecting and cutting may have caused a reduction. Nevertheless, if the magnetic properties were uniform a much greater constancy for R/k would be expected. Accordingly it would appear that the more rapid cooling on the flanks has resulted in properties differing from those in the centre where cooling

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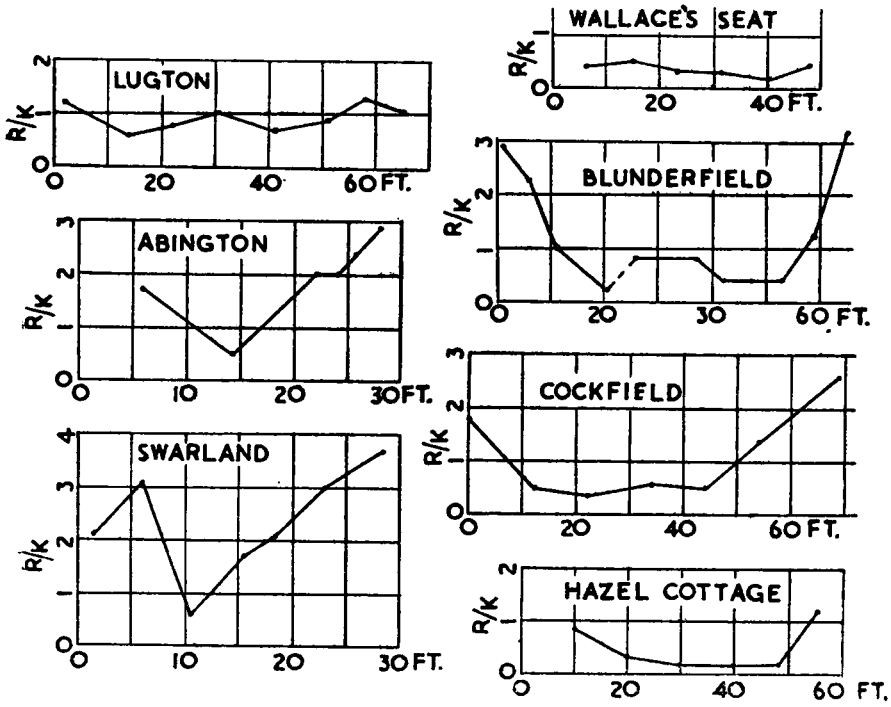


FIG. 9.—*Ratio of residual magnetism to susceptibility.*

TABLE I

Dyke	Exposure	Mean value of R/k at Central Section	Flanks
Acklington	Lugton	0·8	1·2
	Abington	0·5	2·9
	Swarland	1·1	3·0
Cleveland	Wallace's Seat	0·3	0·4
	Blunderfield	0·6	3·1
	Cockfield	0·5	2·2
	Hazel Cottage	0·3	1·2

was slower. Whether this difference arises from chemical composition, crystal structure or grain size is not known.

A very definite change in residual magnetism was observed with time, the specimens being stored approximately in their natural field position in relation to the magnetic field. In general the loss was greatest in those exhibiting strong north polarity on their upper face. Over a period of three months one such specimen showed a 15 per cent decrease in intensity of magnetization, but with the majority it was about 5 per cent. In one peculiar case, from near the centre of the Blunderfield exposure, the value changed from 2200×10^{-6} to 340×10^{-6} c.g.s. units. If certain of the rocks have a small coercive force, then they would be demagnetized in the field in a similar manner. Such characteristics would give a ready explanation of the few cases where normal magnetization occurred. It is unfortunate that, where normal field curves were obtained, no exposures occurred to test this point.

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The authors would like to extend their thanks to the Anglo-Iranian Oil Company, whose financial assistance rendered this work possible. They also acknowledge gratefully the courtesy of the many landowners and tenants who willingly gave them permission to carry out measurements on their property.

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1948 *June* 29.

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